

HW2

Sam Olson

Q1

Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows Model II.

(a)

If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both estimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily estimable? Explain your answer.

Answer. Yes.

Recall that $\underline{\lambda}' \underline{\beta}$ is *estimable* if there exists a linear unbiased estimator $c + \underline{a}' \underline{y}$ of it. We use:

- **Lemma 2.1**, which states that under Models II–VII, a linear estimator $c + \underline{a}' \underline{y}$ of $\underline{\lambda}' \underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}' X$.
- **Theorem 2.1**, which states that under Models II–VII, $\underline{\lambda}' \underline{\beta}$ is estimable if and only if $\underline{\lambda}' = \underline{a}' X$ for some $n \times 1$ vector $\underline{a}$, that is, if and only if $\underline{\lambda}' \in \mathcal{R}(X)$.

Since $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are estimable, Theorem 2.1 gives $n \times 1$ vectors $\underline{a}_1$ and $\underline{a}_2$ with $\underline{\lambda}'_1 = \underline{a}'_1 X$ and $\underline{\lambda}'_2 = \underline{a}'_2 X$. Then

$$\underline{\lambda}'_1 + \underline{\lambda}'_2 = (\underline{a}_1 + \underline{a}_2)' X \in \mathcal{R}(X),$$

so $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta} = (\underline{\lambda}_1 + \underline{\lambda}_2)' \underline{\beta}$ is estimable by Theorem 2.1. By Lemma 2.1, $(\underline{a}_1 + \underline{a}_2)' \underline{y}$ is a linear unbiased estimator of it.

This argument is **Corollary 2 to Theorem 2.1**, which states that under Models II–VII, if $\underline{\lambda}'_1 \underline{\beta}, \dots, \underline{\lambda}'_k \underline{\beta}$ are estimable, then any linear combination of them is estimable.

(b)

If $\underline{\lambda}'_1 \underline{\beta}$ and $\underline{\lambda}'_2 \underline{\beta}$ are both nonestimable, is $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta}$ necessarily nonestimable? Explain your answer.

Answer. No. Here is a counterexample.

Take $p = 2$ and $X = \underline{1}_n \underline{1}'_2$, the $n \times 2$ matrix whose every row is $(1, 1)$, so that $y_i = \beta_1 + \beta_2 + \epsilon_i$. For any $n \times 1$ vector $\underline{a}$, $\underline{a}' X = (\underline{a}' \underline{1}_n) \underline{1}'_2$, so

$$\mathcal{R}(X) = \{t \underline{1}'_2 : t \in \mathbb{R}\} = \{(t, t) : t \in \mathbb{R}\}.$$

Let $\underline{\lambda}_1 = (1, 0)'$ and $\underline{\lambda}_2 = (0, 1)'$. Then $\underline{\lambda}'_1 \notin \mathcal{R}(X)$ and $\underline{\lambda}'_2 \notin \mathcal{R}(X)$, so by Theorem 2.1, $\underline{\lambda}'_1 \underline{\beta} = \beta_1$ and $\underline{\lambda}'_2 \underline{\beta} = \beta_2$ are both nonestimable. However,

$$\underline{\lambda}'_1 + \underline{\lambda}'_2 = (1, 1) \in \mathcal{R}(X),$$

so $\underline{\lambda}'_1 \underline{\beta} + \underline{\lambda}'_2 \underline{\beta} = \beta_1 + \beta_2$ is estimable.

The same conclusion holds in general. If $\underline{\lambda}' \underline{\beta}$ is nonestimable, then $-\underline{\lambda}' \underline{\beta}$ is nonestimable too; otherwise $\underline{\lambda}' \underline{\beta} = (-1)(-\underline{\lambda}' \underline{\beta})$ would be estimable by Corollary 2 to Theorem 2.1. Yet $\underline{\lambda}' \underline{\beta} + (-\underline{\lambda}' \underline{\beta}) = \underline{0}' \underline{\beta}$ is estimable, since $\underline{0}'_p = \underline{0}'_n X \in \mathcal{R}(X)$.

(c)

Can $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ be estimable if $\underline{\lambda}'_1\beta$ is estimable and $\underline{\lambda}'_2\beta$ is nonestimable? Explain your answer.

Answer. No. Suppose, for contradiction, that $\underline{\lambda}'_1\beta$ and $\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta$ are both estimable. Then

$$\underline{\lambda}'_2\beta = (1)(\underline{\lambda}'_1\beta + \underline{\lambda}'_2\beta) + (-1)(\underline{\lambda}'_1\beta)$$

is a linear combination of estimable functions, and so it is estimable by Corollary 2 to Theorem 2.1. This contradicts the assumption that $\underline{\lambda}'_2\beta$ is nonestimable.

Q2

Take the model to be Model VII and suppose that

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

(a)

Find the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and on $\mathcal{C}^\perp(X)$.

Answer. Recall two definitions:

- $\mathcal{C}(X) = \{X\underline{b} : \underline{b} \in \mathbb{R}^3\}$ is the span of the columns of X .
- $\mathcal{C}^\perp(X) = \{\underline{a} \in \mathbb{R}^3 : X'\underline{a} = \underline{0}\}$ is the null space of X' .

We use **Theorem 2.6**, which states that any $n \times 1$ vector $\underline{y}$ admits a unique decomposition $\underline{y} = \underline{y}_1 + \underline{y}_2$ with $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$. It also states that this decomposition is given by $\underline{y}_1 = X\hat{\underline{\beta}}$ and $\underline{y}_2 = \underline{y} - X\hat{\underline{\beta}}$ for any solution $\hat{\underline{\beta}}$ to the normal equations. The vectors $\underline{y}_1$ and $\underline{y}_2$ are the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$, respectively.

Finding $\mathcal{C}(X)$. The columns of X are

$$\underline{x}_1 = (1, 2, 0)', \quad \underline{x}_2 = (2, 3, 0)', \quad \underline{x}_3 = (3, 1, 0)'.$$

Each has third coordinate 0, so $\mathcal{C}(X) \subseteq \{(u, v, 0)' : u, v \in \mathbb{R}\}$. For the reverse inclusion, note that

$$-3\underline{x}_1 + 2\underline{x}_2 = (1, 0, 0)' \quad \text{and} \quad 2\underline{x}_1 - \underline{x}_2 = (0, 1, 0)'$$

both lie in $\mathcal{C}(X)$. Every $(u, v, 0)'$ is a linear combination of these two vectors. Hence

$$\mathcal{C}(X) = \{(u, v, 0)' : u, v \in \mathbb{R}\}.$$

Finding $\mathcal{C}^\perp(X)$. For $\underline{a} = (a_1, a_2, a_3)'$,

$$X'\underline{a} = (a_1 + 2a_2, 2a_1 + 3a_2, 3a_1 + a_2)'.$$

The first two coordinates vanish only if $a_1 = a_2 = 0$, because $\det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = -1 \neq 0$. Then the third coordinate vanishes as well, and a_3 is unrestricted. Hence

$$\mathcal{C}^\perp(X) = \{(0, 0, w)' : w \in \mathbb{R}\}.$$

The projections. Write

$$\underline{y} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}.$$

The first vector is in $\mathcal{C}(X)$ and the second is in $\mathcal{C}^\perp(X)$. By the uniqueness in Theorem 2.6, the orthogonal projections of $\underline{y}$ on $\mathcal{C}(X)$ and $\mathcal{C}^\perp(X)$ are $(2, 3, 0)'$ and $(0, 0, 4)'$, respectively.

(b)

Determine $\text{rank}(X'X)$.

Answer. From part (a), $\mathcal{C}(X)$ is spanned by the two linearly independent vectors $(1, 0, 0)'$ and $(0, 1, 0)'$. So $\text{rank}(X) = \dim \mathcal{C}(X) = 2$.

We use **Lemma 2.3**, which states that for any matrix A , $\text{rank}(A'A) = \text{rank}(A)$. With $A = X$, it gives

$$\text{rank}(X'X) = \text{rank}(X) = 2.$$

(c)

Use the following method and find a generalized inverse of $X'X$.

Let A be an $n \times m$ matrix of rank $r \geq 1$. Let B be a nonsingular $r \times r$ submatrix of A . For matrix A , replace the entries of B by those of $(B^{-1})'$, and replace all other entries of A by 0. Then the transpose of the resulting matrix is a generalized inverse of A .

Answer. Recall that a *generalized inverse* of a matrix A is any matrix G satisfying $AGA = A$.

We have

$$A = X'X = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix}.$$

This 3×3 matrix has rank $r = 2$ by part (b). Take B to be the leading 2×2 submatrix. Then

$$B = \begin{pmatrix} 5 & 8 \\ 8 & 13 \end{pmatrix}, \quad \det(B) = 65 - 64 = 1 \neq 0, \quad (B^{-1})' = B^{-1} = \begin{pmatrix} 13 & -8 \\ -8 & 5 \end{pmatrix},$$

where $(B^{-1})' = B^{-1}$ because B is symmetric. Replacing the entries of B in A by those of $(B^{-1})'$ and all other entries by 0 gives a symmetric matrix, so transposing leaves it unchanged. We obtain

$$G = (X'X)^- = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Check. We have

$$AG = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -7 & 5 & 0 \end{pmatrix} \quad \text{and} \quad AGA = \begin{pmatrix} 5 & 8 & 5 \\ 8 & 13 & 9 \\ 5 & 9 & 10 \end{pmatrix} = A,$$

so G is a generalized inverse of $X'X$.

(d)

Compute a solution $\hat{\underline{\beta}}$ to the normal equations.

Answer. We use two results:

- **Theorem 2.3**, which states that the normal equations $X'X\hat{\underline{\beta}} = X'\underline{y}$ are consistent.
- **Theorem 1.2**, which states that a matrix G is a generalized inverse of a matrix A if and only if $\underline{w} = G\underline{c}$ solves $A\underline{w} = \underline{c}$ for every vector $\underline{c}$ for which these equations are consistent.

Applying Theorem 1.2 with $A = X'X$ and $\underline{c} = X'\underline{y}$, the vector $\hat{\underline{\beta}} = (X'X)^- X'\underline{y}$ solves the normal equations. With

$$X'\underline{y} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 0 \\ 3 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix},$$

we get

$$\hat{\underline{\beta}} = \begin{pmatrix} 13 & -8 & 0 \\ -8 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 8 \\ 13 \\ 9 \end{pmatrix} = \begin{pmatrix} 104 - 104 \\ -64 + 65 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

As a direct check, $X'X\hat{\underline{\beta}}$ is the second column of $X'X$, namely $(8, 13, 9)' = X'\underline{y}$.

(e)

Check that the vectors $X\hat{\underline{\beta}}$ and $\underline{y} - X\hat{\underline{\beta}}$ agree with your results in part (a).

Answer. From part (d), $X\hat{\underline{\beta}}$ is the second column of X , so

$$X\hat{\underline{\beta}} = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix}, \quad \underline{y} - X\hat{\underline{\beta}} = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}.$$

These agree with the projections found in part (a), as Theorem 2.6 requires.

```
X <- matrix(c(1, 2, 3,
              2, 3, 1,
              0, 0, 0), nrow = 3, byrow = TRUE)
y <- c(2, 3, 4)
XtX <- crossprod(X)
G <- matrix(c(13, -8, 0,
              -8, 5, 0,
              0, 0, 0), nrow = 3, byrow = TRUE)

all.equal(XtX %*% G %*% XtX, XtX) # G is a generalized inverse of X'X

## [1] TRUE

bhat <- drop(G %*% crossprod(X, y))
bhat

## [1] 0 1 0

drop(X %*% bhat) # projection onto C(X)

## [1] 2 3 0

drop(y - X %*% bhat) # projection onto C-perp(X)

## [1] 0 0 4
```

Q3

For Model VII, partition $\underline{y}$, X , and $\underline{\epsilon}$ correspondingly as follows:

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \text{and} \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix}.$$

Suppose that the row spaces of X_1 and X_2 are essentially disjoint, that is, $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{\underline{0}'\}$. Show that, if $\underline{\lambda}' \in \mathcal{R}(X_1)$, then the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y}_1 = X_1\underline{\beta} + \underline{\epsilon}_1$ is the same as the least squares estimate of $\underline{\lambda}'\underline{\beta}$ based on the model $\underline{y} = X\underline{\beta} + \underline{\epsilon}$.

Hint: Under the full model, the LSE is $\underline{a}'X'\underline{y} = \underline{a}'X'_1\underline{y}_1 + \underline{a}'X'_2\underline{y}_2$ and, under the partial model, the LSE is $\underline{b}'X'_1\underline{y}_1$ where here $X'_1X_1\underline{a} + X'_2X_2\underline{a} = X'X\underline{a} = \underline{\lambda} = X'_1X_1\underline{b}$ (why?); in the last equation, move terms involving X'_2 and X'_1 to different sides.

Answer.

Setup. Let X_1 be $n_1 \times p$ and X_2 be $n_2 \times p$, with $n_1 + n_2 = n$. The hint's $\underline{a}$ and $\underline{b}$ are $p \times 1$ solutions to conjugate normal equations. To keep $\underline{a}$ reserved for $n \times 1$ coefficient vectors, we write them as $\underline{r}$ and $\underline{r}_1$. The partial model $\underline{y}_1 = X_1 \underline{\beta} + \underline{\epsilon}_1$ is itself an instance of Model VII: $E(\underline{\epsilon}_1) = \underline{0}_{n_1}$, and $\text{Var}(\underline{\epsilon}_1)$ is the leading $n_1 \times n_1$ block of $\text{Var}(\underline{\epsilon})$. So every Model II–VII result below applies to both models.

Estimability under both models. Since $\underline{\lambda}' \in \mathcal{R}(X_1)$, we have $\underline{\lambda}' = \underline{t}' X_1$ for some $n_1 \times 1$ vector $\underline{t}$. Hence

$$\underline{\lambda}' = (\underline{t}', \underline{0}'_{n_2}) \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \in \mathcal{R}(X).$$

By Theorem 2.1 (stated in Q1(a)), $\underline{\lambda}' \underline{\beta}$ is estimable under both models.

Theorem 2.5 states that under Models II–VII, $\underline{\lambda}' \hat{\underline{\beta}} = \underline{\lambda}' \tilde{\underline{\beta}}$ for any two solutions $\hat{\underline{\beta}}$ and $\tilde{\underline{\beta}}$ to the normal equations if and only if $\underline{\lambda}' \underline{\beta}$ is estimable. So the least squares estimate of $\underline{\lambda}' \underline{\beta}$ is well defined under *each* model. However, the two models have different normal equations, so this alone does not show that the two estimates coincide. That is what remains to prove.

Representing the two estimates (the “why?” in the hint). We use two results:

- **Theorem 2.7**, which states that under Models II–VII, $\underline{\lambda}' \underline{\beta}$ is estimable if and only if the conjugate normal equations $X' X \underline{r} = \underline{\lambda}$ are consistent.
- **Theorem 2.8**, which states that under Models II–VII, the least squares estimate of an estimable $\underline{\lambda}' \underline{\beta}$ equals $\underline{r}' X' \underline{y}$, where $\underline{r}$ is any solution to $X' X \underline{r} = \underline{\lambda}$.

Since $\underline{\lambda}' \underline{\beta}$ is estimable under both models, Theorem 2.7 gives $p \times 1$ vectors $\underline{r}$ and $\underline{r}_1$ with

$$X' X \underline{r} = \underline{\lambda} \quad \text{and} \quad X_1' X_1 \underline{r}_1 = \underline{\lambda}.$$

By Theorem 2.8, the least squares estimates under the full and partial models are $\underline{r}' X' \underline{y}$ and $\underline{r}_1' X_1' \underline{y}_1$, respectively. By block multiplication,

$$X' X = X_1' X_1 + X_2' X_2 \quad \text{and} \quad X' \underline{y} = X_1' \underline{y}_1 + X_2' \underline{y}_2.$$

Therefore

$$X_1' X_1 \underline{r} + X_2' X_2 \underline{r} = \underline{\lambda} = X_1' X_1 \underline{r}_1, \quad \text{that is,} \quad X_1' X_1 (\underline{r} - \underline{r}_1) = -X_2' X_2 \underline{r}.$$

Using disjointness. Transposing both sides gives an equality of $1 \times p$ row vectors:

$$\underbrace{[X_1(\underline{r} - \underline{r}_1)]' X_1}_{\in \mathcal{R}(X_1)} = \underbrace{[-X_2 \underline{r}]' X_2}_{\in \mathcal{R}(X_2)}.$$

The left side has the form $\underline{u}' X_1$ with $\underline{u} = X_1(\underline{r} - \underline{r}_1) \in \mathbb{R}^{n_1}$, so it lies in $\mathcal{R}(X_1)$. Similarly, the right side lies in $\mathcal{R}(X_2)$. Since $\mathcal{R}(X_1) \cap \mathcal{R}(X_2) = \{\underline{0}'\}$, both sides equal $\underline{0}'_p$. This gives two facts.

1. $\underline{r}' X_2' X_2 = \underline{0}'_p$. Hence $(X_2 \underline{r})' (X_2 \underline{r}) = \underline{r}' X_2' X_2 \underline{r} = 0$. We use **Lemma 1.2**, which states that for any matrix M , $\text{tr}(M' M) = 0$ implies $M = 0$. Applied to the $n_2 \times 1$ matrix $M = X_2 \underline{r}$, for which $\text{tr}(M' M) = (X_2 \underline{r})' (X_2 \underline{r}) = 0$, it gives $X_2 \underline{r} = \underline{0}_{n_2}$.
2. $X_1' X_1 \underline{r} = X_1' X_1 \underline{r}_1$. We use **Lemma 1.3(ii)**, which states that for any matrices M , P , and Q of appropriate dimensions, $M' M P = M' M Q$ implies $M P = M Q$. Applied with $M = X_1$, $P = \underline{r}$, and $Q = \underline{r}_1$, it gives $X_1 \underline{r} = X_1 \underline{r}_1$.

Conclusion. Combining these,

$$\underline{r}' X' \underline{y} = \underline{r}' X_1' \underline{y}_1 + \underline{r}' X_2' \underline{y}_2 = (X_1 \underline{r})' \underline{y}_1 + (X_2 \underline{r})' \underline{y}_2 = (X_1 \underline{r}_1)' \underline{y}_1 + 0 = \underline{r}_1' X_1' \underline{y}_1.$$

So the least squares estimates of $\underline{\lambda}' \underline{\beta}$ under the two models coincide. $\square$

Q4

Show that, under Model II, $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$.

Answer. Here $\underline{a}$ is $n \times 1$. Under Model II, $E(\underline{\epsilon}) = \underline{0}_n$ and $\text{Var}(\underline{y}) = \text{Var}(\underline{\epsilon}) = \sigma^2 I_n$ with $\sigma^2 > 0$.

The projection matrix. Let $P_X = X(X'X)^{-}X'$, an $n \times n$ matrix. We use three results:

- **Lemma 1.4**, which states that for any matrix X , $X'X(X'X)^{-}X' = X'$.
- **Corollary 1(ii) to Lemma 1.4**, which states that for any matrix X , $X(X'X)^{-}X'X = X$.
- **Corollary 3 to Lemma 1.5**, which states that for any matrix X , $X(X'X)^{-}X'$ is symmetric.

These give three properties of P_X :

- $P_X X = X$, by Corollary 1(ii) to Lemma 1.4.
- $P_X' = P_X$, by Corollary 3 to Lemma 1.5.
- $P_X^2 = X(X'X)^{-}[X'X(X'X)^{-}X'] = X(X'X)^{-}X' = P_X$, by Lemma 1.4.

Consequently,

$$P_X'(I_n - P_X) = P_X - P_X^2 = 0_{n \times n}.$$

The BLUE of $E(\underline{a}'\underline{y})$. We have $E(\underline{a}'\underline{y}) = \underline{a}'X\beta = \underline{\lambda}'\beta$ with $\underline{\lambda}' = \underline{a}'X \in \mathcal{R}(X)$. By Lemma 2.1 (stated in Q1(a)), $\underline{a}'\underline{y}$ is a linear unbiased estimator of $\underline{\lambda}'\beta$, and $\underline{\lambda}'\beta$ is estimable.

We use **Theorem 2.9 (Gauss–Markov)**, which states the following. Under Models II–VII, if $\underline{\lambda}'\beta$ is estimable and $\hat{\beta}$ is any solution to the normal equations, then $\underline{\lambda}'\hat{\beta}$ is a linear unbiased estimator of $\underline{\lambda}'\beta$. Moreover, under Model II, the variance of $\underline{\lambda}'\hat{\beta}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\beta$, so $\underline{\lambda}'\hat{\beta}$ is the BLUE.

By Theorems 2.3 and 1.2 (stated in Q2(d)), $\hat{\beta} = (X'X)^{-}X'\underline{y}$ solves the normal equations. So the BLUE of $\underline{\lambda}'\beta$ is

$$\underline{\lambda}'\hat{\beta} = \underline{a}'X(X'X)^{-}X'\underline{y} = \underline{a}'P_X\underline{y} = (P_X\underline{a})'\underline{y},$$

where the last step uses $P_X' = P_X$.

Comparing variances. We use the **Corollary to Lemma 2.4**, which states that $\text{Var}(L'\underline{x}) = L'\Sigma L$ for a random vector $\underline{x}$ with $\text{Var}(\underline{x}) = \Sigma$ and a matrix of constants L .

Write $\underline{a} = P_X\underline{a} + (I_n - P_X)\underline{a}$. Applying the corollary with $L = \underline{a}$ and $\Sigma = \sigma^2 I_n$,

$$\text{Var}(\underline{a}'\underline{y}) = \sigma^2 \underline{a}'\underline{a} = \sigma^2 (P_X\underline{a})'(P_X\underline{a}) + 2\sigma^2 \underline{a}'P_X'(I_n - P_X)\underline{a} + \sigma^2 [(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}].$$

The middle term vanishes because $P_X'(I_n - P_X) = 0_{n \times n}$. The first term is $\text{Var}((P_X\underline{a})'\underline{y})$, again by the Corollary to Lemma 2.4. Hence

$$\text{Var}(\underline{a}'\underline{y}) = \text{Var}((P_X\underline{a})'\underline{y}) + \sigma^2 [(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}]. \quad (\text{Q4.1})$$

Characterizing when $\underline{a}'\underline{y}$ is a BLUE. Since $\underline{a}'\underline{y}$ is linear and unbiased for $\underline{\lambda}'\beta$, and $(P_X\underline{a})'\underline{y}$ is the BLUE, $\underline{a}'\underline{y}$ is a BLUE if and only if $\text{Var}(\underline{a}'\underline{y}) = \text{Var}((P_X\underline{a})'\underline{y})$. Because $\sigma^2 > 0$, (Q4.1) shows this holds if and only if $[(I_n - P_X)\underline{a}]'[(I_n - P_X)\underline{a}] = 0$. By Lemma 1.2 (stated in Q3), applied to the $n \times 1$ matrix $(I_n - P_X)\underline{a}$, this holds if and only if $(I_n - P_X)\underline{a} = \underline{0}_n$, that is, $\underline{a} = P_X\underline{a}$.

Characterizing $\underline{a} = P_X\underline{a}$. Recall that $\underline{a} \in \mathcal{C}(X)$ if and only if $\underline{a} = X\underline{b}$ for some $p \times 1$ vector $\underline{b}$.

- ($\Rightarrow$) If $\underline{a} = P_X\underline{a}$, then $\underline{a} = X\underline{b}$ with $\underline{b} = (X'X)^{-}X'\underline{a}$, so $\underline{a} \in \mathcal{C}(X)$.
- ($\Leftarrow$) If $\underline{a} = X\underline{b}$, then $P_X\underline{a} = P_X X\underline{b} = X\underline{b} = \underline{a}$, since $P_X X = X$.

Hence $\underline{a}'\underline{y}$ is a BLUE of its expectation if and only if $\underline{a} \in \mathcal{C}(X)$. $\square$

Q5

(a)

Show that, under Model II, the problem of determining the BLUE $\underline{a}'\underline{y}$ of an estimable function $\underline{\lambda}'\underline{\beta}$ is equivalent to that of minimizing $\underline{a}'\underline{a}$ subject to the constraints $\underline{a}'X = \underline{\lambda}'$.

Answer. Here $\underline{a}$ is $n \times 1$ and $\underline{\lambda}$ is $p \times 1$.

The class of linear unbiased estimators. By Lemma 2.1 (stated in Q1(a)), a linear estimator $c + \underline{a}'\underline{y}$ is unbiased for $\underline{\lambda}'\underline{\beta}$ if and only if $c = 0$ and $\underline{a}'X = \underline{\lambda}'$. So the linear unbiased estimators of $\underline{\lambda}'\underline{\beta}$ are exactly

$$\{\underline{a}'\underline{y} : \underline{a}'X = \underline{\lambda}'\},$$

which is nonempty because $\underline{\lambda}'\underline{\beta}$ is estimable.

Their variances. Under Model II, $\text{Var}(\underline{y}) = \sigma^2 I_n$. By the Corollary to Lemma 2.4 (stated in Q4),

$$\text{Var}(\underline{a}'\underline{y}) = \underline{a}'(\sigma^2 I_n)\underline{a} = \sigma^2 \underline{a}'\underline{a}.$$

Equivalence. Since $\sigma^2 > 0$ does not depend on $\underline{a}$, a vector $\underline{a}$ minimizes $\text{Var}(\underline{a}'\underline{y})$ over this class if and only if it minimizes $\underline{a}'\underline{a}$ subject to $\underline{a}'X = \underline{\lambda}'$ (equivalently, $X'\underline{a} = \underline{\lambda}$). The minimizing $\underline{a}$ does not depend on σ^2 , so the resulting estimator has minimum variance uniformly in σ^2 . $\square$

(b)

The Lagrangian function associated with this minimization problem is

$$L(\underline{a}, \underline{\rho}) = \underline{a}'\underline{a} + 2\underline{\rho}'(X'\underline{a} - \underline{\lambda}).$$

Use the vector calculus to differentiate the Lagrangian function with respect to $\underline{a}$ and the Lagrangian multipliers, and set these derivatives equal to zero.

Answer. There is one multiplier per constraint in $X'\underline{a} = \underline{\lambda}$, so $\underline{\rho}$ is $p \times 1$.

Vector-calculus identities. We use the identities in Section 2.3 of the notes, which state the following for a vector variable $\underline{\beta}$:

- For any constant vector $\underline{h}$, $\partial(\underline{h}'\underline{\beta})/\partial\underline{\beta} = \underline{h}$.
- For any constant matrix X , $\partial(\underline{\beta}'X'\underline{X}\underline{\beta})/\partial\underline{\beta} = 2X'X\underline{\beta}$.

Derivative with respect to $\underline{a}$.

- The second identity, with I_n in place of X and $\underline{a}$ in place of $\underline{\beta}$, gives $\partial(\underline{a}'\underline{a})/\partial\underline{a} = 2\underline{a}$.
- Writing $\underline{\rho}'X'\underline{a} = (X\underline{\rho})'\underline{a}$, the first identity with $\underline{h} = X\underline{\rho}$ gives $\partial(\underline{\rho}'X'\underline{a})/\partial\underline{a} = X\underline{\rho}$.
- The term $-2\underline{\rho}'\underline{\lambda}$ does not involve $\underline{a}$.

Derivative with respect to $\underline{\rho}$. Writing $2\underline{\rho}'(X'\underline{a} - \underline{\lambda}) = 2(X'\underline{a} - \underline{\lambda})'\underline{\rho}$, the first identity with $\underline{h} = X'\underline{a} - \underline{\lambda}$ applies.

Together, the $n \times 1$ and $p \times 1$ gradients are

$$\frac{\partial L}{\partial \underline{a}} = 2\underline{a} + 2X\underline{\rho}, \quad \frac{\partial L}{\partial \underline{\rho}} = 2(X'\underline{a} - \underline{\lambda}).$$

Setting these equal to zero gives the system

$$\underline{a} + X\underline{\rho} = \underline{0}_n, \quad X'\underline{a} = \underline{\lambda}. \tag{1}$$

(c)

Show that the resulting equations have a solution if and only if the conjugate normal equations are consistent.

Answer.

($\Rightarrow$) Suppose $(\underline{a}_0, \underline{\rho}_0)$ solves (1). The first equation gives $\underline{a}_0 = -X\underline{\rho}_0$. Substituting into the second,

$$\underline{\lambda} = X'\underline{a}_0 = X'X(-\underline{\rho}_0).$$

Thus $\underline{r} = -\underline{\rho}_0$ solves $X'X\underline{r} = \underline{\lambda}$, so the conjugate normal equations are consistent.

($\Leftarrow$) Suppose $X'X\underline{r}_0 = \underline{\lambda}$ for some $p \times 1$ vector $\underline{r}_0$. Set $\underline{a}_0 = X\underline{r}_0$ and $\underline{\rho}_0 = -\underline{r}_0$. Then

$$\underline{a}_0 + X\underline{\rho}_0 = X\underline{r}_0 - X\underline{r}_0 = \underline{0}_n \quad \text{and} \quad X'\underline{a}_0 = X'X\underline{r}_0 = \underline{\lambda},$$

so $(\underline{a}_0, \underline{\rho}_0)$ solves (1). □

(d)

Solve your equations for the minimizing value of $\underline{a}$.

Answer.

Existence of a solution. Since $\underline{\lambda}'\underline{\beta}$ is estimable, the conjugate normal equations are consistent by Theorem 2.7 (stated in Q3). So (1) has a solution by part (c).

All solutions. The first equation of (1) gives $\underline{a} = -X\underline{\rho}$. Substituting into the second gives $X'X(-\underline{\rho}) = \underline{\lambda}$, a consistent system in the $p \times 1$ unknown $-\underline{\rho}$.

We use **Theorem 1.3**, which states that all solutions to a consistent matrix equation $AW = Y$ are generated by

$$W = A^-Y + (I - A^-A)Z$$

as Z ranges over all matrices of appropriate dimensions. (We write W for the notes' unknown to avoid a clash with the design matrix X .) Applying it with $A = X'X$, $W = -\underline{\rho}$, and $Y = \underline{\lambda}$, all solutions are

$$-\underline{\rho} = (X'X)^-\underline{\lambda} + [I_p - (X'X)^-X'X]\underline{z}, \quad \underline{z} \in \mathbb{R}^p.$$

Hence

$$\underline{a} = -X\underline{\rho} = X(X'X)^-\underline{\lambda} + [X - X(X'X)^-X'X]\underline{z} = X(X'X)^-\underline{\lambda},$$

since $X(X'X)^-X'X = X$ by Corollary 1(ii) to Lemma 1.4 (stated in Q4). Therefore every solution of (1) has the same $\underline{a}$ -component,

$$\underline{a}_* = X(X'X)^-\underline{\lambda}.$$

The stationary point is the minimum. Equations (1) are necessary conditions only, so we verify directly that $\underline{a}_*$ minimizes $\underline{a}'\underline{a}$ subject to $X'\underline{a} = \underline{\lambda}$.

Let $\underline{a}$ be any $n \times 1$ vector with $X'\underline{a} = \underline{\lambda}$. Set $\underline{d} = \underline{a} - \underline{a}_*$; then $X'\underline{d} = \underline{\lambda} - \underline{\lambda} = \underline{0}_p$. Since $\underline{a}_* = X\underline{r}_*$ with $\underline{r}_* = (X'X)^-\underline{\lambda}$, we have

$$\underline{a}'_*\underline{d} = \underline{r}'_*X'\underline{d} = 0.$$

Therefore

$$\underline{a}'\underline{a} = (\underline{a}_* + \underline{d})'(\underline{a}_* + \underline{d}) = \underline{a}'_*\underline{a}_* + \underline{d}'\underline{d} \geq \underline{a}'_*\underline{a}_*.$$

Equality holds if and only if $\underline{d}'\underline{d} = 0$, that is, $\underline{d} = \underline{0}_n$ (by Lemma 1.2, stated in Q3). So $\underline{a}_*$ is the unique minimizer.

The resulting BLUE. By part (a), the BLUE is

$$\underline{a}'_*\underline{y} = \underline{\lambda}'[(X'X)^-]'X'\underline{y}.$$

We use the **Corollary to Lemma 1.1**, which states that for any matrix X , $[(X'X)^-]'$ is a generalized inverse of $X'X$. By Theorems 2.3 and 1.2 (stated in Q2(d)), $\widehat{\underline{\beta}} = [(X'X)^-]'X'\underline{y}$ therefore solves the normal equations, and

$$\underline{a}'_*\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}}.$$

That is, the BLUE of $\underline{\lambda}'\underline{\beta}$ is its least squares estimator, in agreement with Theorem 2.9 (stated in Q4). $\square$